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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Swarm Robotics based Multi-Agent Coordination System

CSA1709-Artificial Intelligence

Presented by:

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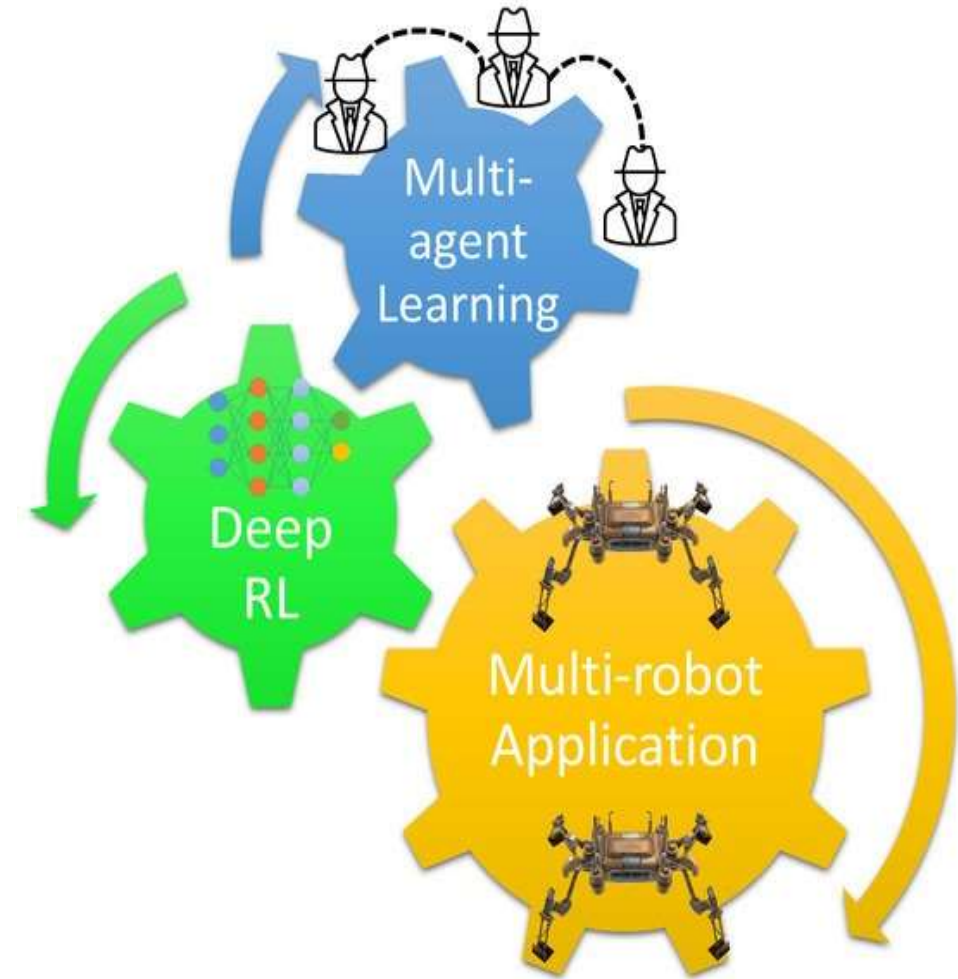
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INTRODUCTION

- Swarm Robotics is a technology in which multiple robots work together to perform a common task.
- Each robot acts as an independent agent and communicates with other robots.
- The system is inspired by the collective behaviour of ants, bees, and birds.
- Multi-agent coordination allows robots to share information, divide tasks, and make decisions collectively.



OBJECTIVES

- To develop a system where multiple robots work together to complete a common task.
- To enable communication and information sharing between robots.
- To implement efficient task allocation and coordination among multiple agents.
- To achieve collision-free movement and obstacle avoidance.
- To improve area coverage and task completion time.
- To enable robots to make distributed and autonomous decisions.



PROBLEM STATEMENT



- Develop a Swarm Robotics based Multi-Agent Coordination System.
- Enable multiple robots to work together autonomously.
- Provide robot-to-robot communication for sharing information.
- Implement efficient task allocation among robots.
- Enable collision and obstacle avoidance.
- Support real-time decision-making in dynamic environments.
- Improve area coverage and task completion efficiency.



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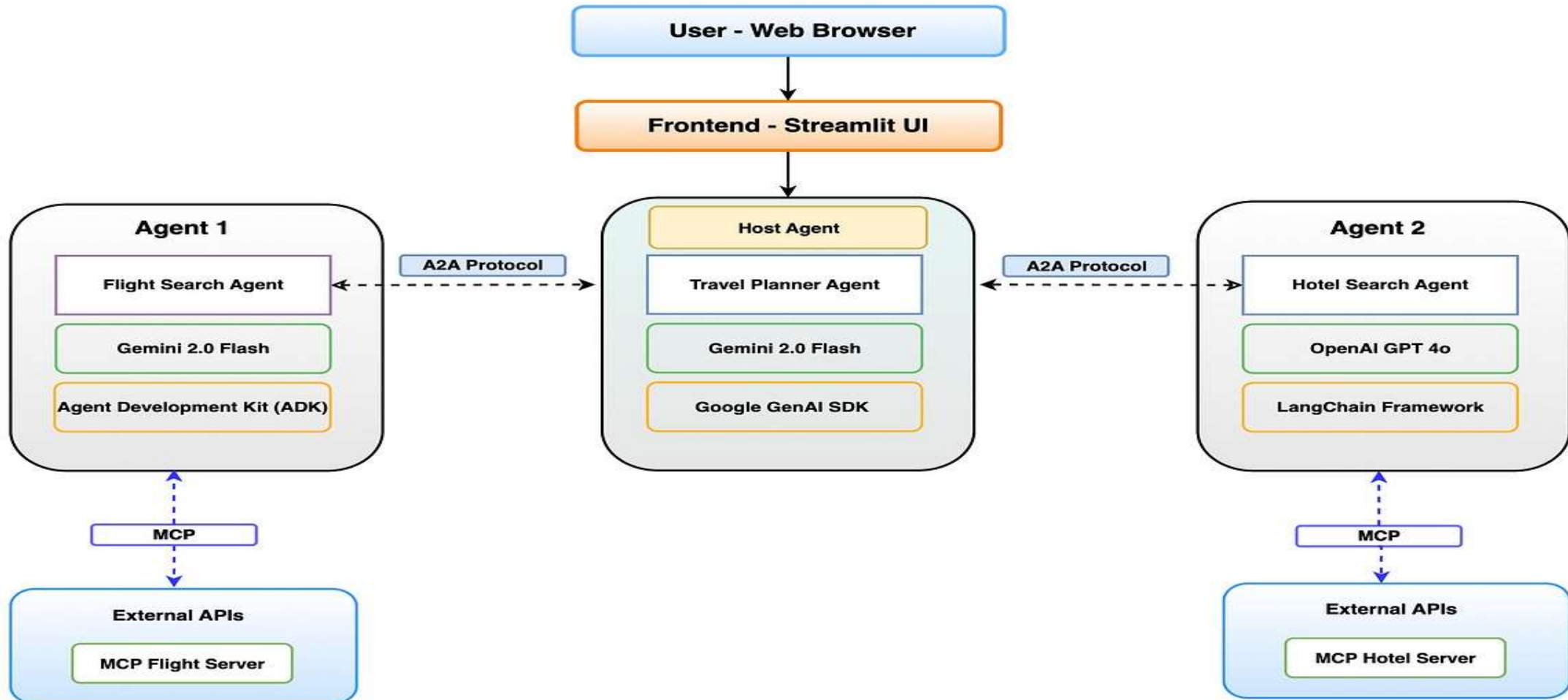


Literature Review Background

S. No.	Author(s) & Year	Title / Reference	Research Focus	Method / Approach	Key Findings / Contributions	Limitations	Relevance to Proposed System
1	Beni, G. (2004)	<i>Swarm Robotics: From Sources of Inspiration to Domains of Application</i>	General introduction to swarm robotics, principles, and applications.	- Literature survey - Conceptual framework	Defined swarm robotics and discussed inspiration from natural systems and major application domains.	Conceptual paper, no algorithm or implementation.	Provides foundational concepts and inspiration for multi-agent coordination.
2	Trianni, V. (2008)	<i>Evolutionary Robotics and Swarm Intelligence</i>	Use of evolutionary algorithms and swarm intelligence in robotics.	- Evolutionary algorithms - Swarm models	Showed how evolution and swarm intelligence improve adaptability and robustness of robot swarms.	High computational cost; requires simulation tuning.	Supports use of swarm intelligence for robust multi-agent coordination.
3	Sahin, E. (2005)	<i>Swarm Robotics: From Sources of Inspiration to Domains of Application</i>	Review of biological inspiration and swarm robotic behaviors.	- Bio-inspired methods - Behavior-based control	Identified key swarm behaviors such as aggregation, dispersion, and pattern formation.	Mostly theoretical overview.	Helps in designing behavior rules for multi-agent coordination.
4	Kennedy, J., Eberhart, R. (1995)	<i>Particle Swarm Optimization</i>	Optimization using social behavior of particles.	Particle Swarm Optimization (PSO)	PSO is simple, efficient, and suitable for multi-robot, coordination and parameter optimization.	Susceptible to local optima in complex environments.	Can be used to optimize movement, formation, and task allocation.
5	Zhang, L., Muthukrishnan, S. (2019)	<i>Reinforcement Learning in Swarm Robotics</i>	Use of reinforcement learning for adaptive swarm behavior.	- Reinforcement Learning (RL) - Simulation environment	RL enables robots to learn policies for navigation, obstacle avoidance and cooperation.	Requires large training data and computational resources.	Can be extended to make the swarm adaptive and self-learning.

SYSTEM ARCHITECTURE

Multi Agent Travel Planner using A2A , MCP and ADK , LangChain





Module 1

Swarm Coordination and Communication Algorithm



- Enables multiple robots to communicate and work together as a swarm.
- Robots exchange information such as position, direction, sensor data, and task status.
- Uses communication technologies such as Wi-Fi, Bluetooth, or RF modules.
- Implements coordination algorithms to decide which robot performs which task.
- Robots share local information to improve overall swarm performance.
- Includes collision avoidance and coordinated movement.

Basic Working

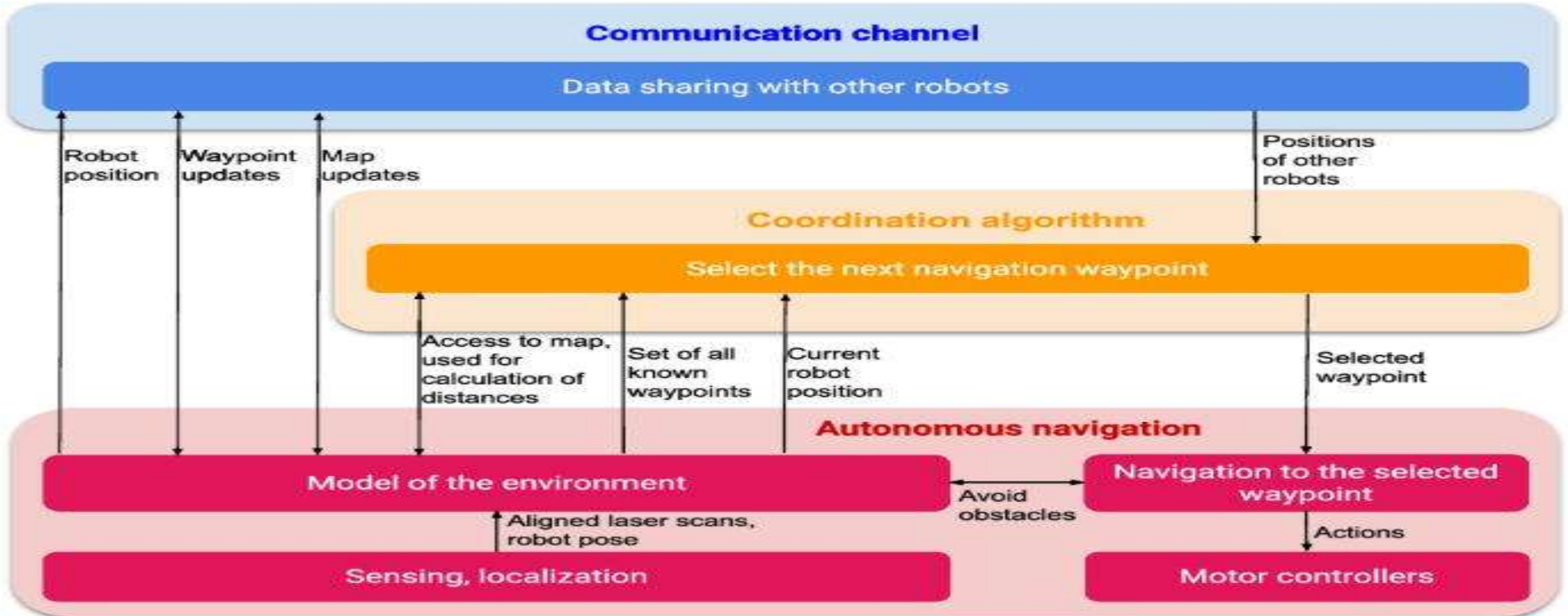
Sense → Communicate → Decide → Coordinate → Move → Repeat



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SYSTEM ARCHITECTURE





Module 2

Task Allocation and Collective Goal Achievement



- Assigns different tasks to individual robots based on their location, capability, and availability.
- The swarm divides a large task into smaller tasks that can be completed simultaneously.
- Robots communicate their task status and progress with other agents.
- If a robot fails or becomes unavailable, its task can be reassigned to another robot.
- Robots coordinate their activities to avoid duplicate work and collisions.
- The system continuously monitors task completion and updates the task allocation



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OUTPUT

SR-MACS

Swarm Robotics Multi-Agent Coordination System

SWARM ROBOTICS

MULTI-AGENT COORDINATION SYSTEM

SYSTEM PAUSED

Swarm Robotics Based Multi-Agent Coordination System

Autonomous Multi-Agent Coordination, Navigation & Task Allocation

▶ START SIMULATION

⏸ PAUSE

↺ RESET

🎲 RANDOMIZE

🛑 EMERGENCY STOP

📄 EXPORT SIMULATION DATA

SWARM CONFIGURATION

Number of Robots 8

Number of Targets 3

Simulation Speed 1x

Communication Range 150 px

Obstacle Density Medium

3 TARGETS

0 LINKS

INTERACTION

Click → Add Obstacle

Shift + Click → Add Target

Robot Click → Robot Details

Double-Click Target → Remove Target

PERFORMANCE METRICS

8

ACTIVE ROBOTS

0

TARGETS COMPLETED

0

COLLISIONS

12

OBSTACLES

0

COMM LINKS

90%

SWARM EFFICIENCY

0.6 m/s

AVERAGE SPEED

0%

COVERAGE



Results & outcomes



- Provided interactive controls including Start, Pause, Reset, Randomize, Emergency Stop, and Export Simulation Data.
- Successfully simulated 8 autonomous robots operating in a shared environment.
- Enabled multi-agent coordination for movement, navigation, and task execution.
- Supported detection and navigation around 12 obstacles.
- Managed 3 target locations for cooperative task completion.
- Monitored important metrics such as active robots, completed targets, collisions, communication links, speed, and coverage.
- Achieved efficient decentralized coordination with reduced collision risk.



CHALLENGES & LIMITATIONS



Challenges:

Communication: Reliable communication between multiple robots can be difficult.

Coordination: Maintaining cooperation among many robots is complex.

Collision Avoidance: Robots must avoid collisions with other robots and obstacles.

Limitations:

- Limited processing and sensing capabilities of individual robots.
- Dependence on wireless communication can cause delays and data loss.
- AI-based coordination may require large training datasets and computational resources.

CONCLUSION

- Enables multiple robots to work cooperatively toward a common goal.
- Uses decentralized control for independent decision-making.
- Improves scalability and flexibility as the number of robots increases.
- Provides effective task allocation among multiple agents.
- Supports collision avoidance and safe navigation.
- Enables formation control and coordinated movement.



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Thank You!